

PRE-MEETING NOTES (11/05)

AGATHA:

licS Paper: P_n and λ_n + hierarchy

full lambda k
calc. with exchange

Thm: axioms in $\lambda_2 \iff$ finite set of structural rules

Thm: axioms in $\beta_3 \iff$ finite set of hyperstructural rules in $\#Flew$

Rule completion

TOCL Paper: 2-systems (generalised geometric implications)

2-systems $\iff$ hypersequent rules

derivations + normal forms

NOTE: the embedding allows for simpler (?)
analytic proofs

relies on the fact that we manipulate
only atoms.

For Hilbert axioms in GA_1 , we can recover
analyticity by using the completion procedure.

NEOWi

JLC 2016 :

GA_0 and GA_n

GA_1 to GRS_1 and GA_{n+1} to GRS_{n+1}

(manipulate only atomic formulas)

Draft (Tzaho Teri)

Generalization for intuitionistic logic
(add constants)

Nested Implications

localizing nested rules

OBSERVATION :

Some issues w/ Aycotha's embedding (?)

From hypersequents to 2-system derivation
there are some really obscure steps...

Sometimes, translating a 2-sequent into a
hypersequent rule creates issues such as:

$$\begin{array}{c}
 \frac{\Gamma \Rightarrow \Delta}{\varphi, \Gamma \Rightarrow \Delta} r_1 \quad \frac{\Gamma'' \Rightarrow \Delta''}{\psi, \Gamma \Rightarrow \Delta} r_2 \\
 \hline
 \varphi \vee \psi, \Gamma \Rightarrow \Delta \quad \text{VL} \\
 \vdots
 \end{array}$$

$$\frac{\Gamma_0 \Rightarrow \Delta_0 \quad \dots \quad \Gamma'_0 \Rightarrow \Delta'_0}{\Gamma_0 \Rightarrow \Delta_0} r$$

this would be because:

$$\frac{G \mid \Gamma' \Rightarrow \Delta' \quad G \mid \Gamma'' \Rightarrow \Delta''}{G \mid \varphi, \Gamma \Rightarrow \Delta \mid \psi, \Gamma \Rightarrow \Delta}$$

It is now impossible to continue translating the derivation, because there is only 1 hp. corresponding to the top rules.

(likewise w/ $\wedge L$ and etc)

$$QS: \quad \forall x (A(x) \vee B) \rightarrow (\forall x A(x) \vee B)$$

gives the 2-systems:

$$\frac{\frac{T', A(t) \vdash c'}{T' \vdash c'} \quad \frac{T'', B \vdash c''}{T'' \vdash c''} \quad \frac{}{T''', A(y)/B \vdash c'''}{\vdots \quad \vdots \quad \vdots} \quad \frac{T \vdash c \quad T \vdash c \quad T \vdash c}{T \vdash c}$$

modulo eigenvariables

would translate to the hypersequent:

$$\frac{G \mid T', A(t) \vdash \pi' \quad G \mid T'', B \vdash c''}{G \mid T' \vdash \pi' \mid T' \vdash \pi'' \mid T''', A(y) \vdash c'''}$$

$$\begin{array}{ccc} \frac{}{A(t), A(y) \vdash A(y)} \text{Id} & & \frac{}{A(t), A(z) \vdash A(z)} \text{Id} \\ \frac{}{A(t) \vdash A(y)} \text{g}_1 & \frac{}{B \vdash A(y)} \text{g}_3 & \frac{}{A(t) \vdash A(z)} \text{g}_1 \quad \frac{}{B \vdash A(z)} \text{g}_3 \\ \frac{A(t) \vdash A(y) \quad B \vdash A(y)}{A(t) \vee B \vdash A(y)} \text{g}_2 & \frac{}{A(t), B \vdash B} \text{Id} & \frac{}{A(t) \vdash A(z)} \text{g}_1 \quad \frac{}{B \vdash A(z)} \text{g}_3 \\ \frac{}{A(t) \vee B \vdash A(y)} \text{g}_2 & \frac{}{A(t) \vdash B} \text{g}_2 & \frac{}{B \vdash B} \text{Id} \\ \frac{}{A(t) \vee B \vdash \forall x A(x)} \text{g}_2 & \frac{}{A(t) \vee B \vdash B} \text{Id} & \frac{}{A(t) \vee B \vdash \forall x A(x)} \text{g}_2 \\ \frac{}{A(t) \vee B \vdash \forall x A(x) \vee B} \text{g}_2 & \frac{}{A(t) \vee B \vdash \forall x A(x) \vee B} \text{g}_2 & \frac{}{A(t) \vee B \vdash \forall x A(x) \vee B} \text{g}_2 \\ \frac{}{\forall x (A(x) \vee B) \vdash \forall x A(x) \vee B} \text{g}_2 & \frac{}{\forall x (A(x) \vee B) \vdash \forall x A(x) \vee B} \text{g}_2 & \frac{}{\forall x (A(x) \vee B) \vdash \forall x A(x) \vee B} \text{g}_2 \\ \frac{}{\vdash \forall x (A(x) \vee B) \supset (\forall x A(x) \vee B)} \text{g}_2 & \frac{}{\vdash \forall x (A(x) \vee B) \supset (\forall x A(x) \vee B)} \text{g}_2 & \frac{}{\vdash \forall x (A(x) \vee B) \supset (\forall x A(x) \vee B)} \text{g}_2 \end{array}$$

EINOVARIABLES:

Example 1: Sere's draft

Example: $\forall x (P(x) \supset \exists y Q(x,y)) \vee \exists u \forall w (Q(u,w) \supset P(w))$ ~~EGA~~

$$\begin{array}{c}
 \frac{\Sigma', y; T'' \Downarrow Q(x,y) \vdash C'}{y'} \\
 \frac{\Sigma', T'' \Downarrow \exists y Q(x,y) \vdash C' \quad \Sigma', T' \vdash P(x) \Downarrow}{\vdash C'} \supset_L \\
 \frac{\Sigma', T' \Downarrow P(x) \supset \exists y Q(x,y) \vdash C'}{\vdash C'} \\
 \vdots \\
 \frac{\Sigma', T' \Downarrow \forall x (P(x) \supset \exists y Q(x,y)) \vdash C'}{\vdash C'} \\
 \vdots \\
 \frac{\Sigma', u; T'' \Downarrow P(t) \vdash C'' \quad \Sigma', u; T'' \vdash Q(u,t)}{\Sigma', u; T'' \Downarrow Q(u,t) \supset P(t) \vdash C''} \\
 \frac{\Sigma', u; T'' \Downarrow Q(u,t) \supset P(t) \vdash C''}{\Sigma', u; T'' \Downarrow \forall () \vdash C''} \\
 \vdots \\
 \frac{u; T' \Downarrow \forall w (Q(u,w) \supset P(w)) \vdash C}{\vdash C} \exists_L \\
 \frac{\vdash \forall x (P(x) \supset \exists y Q(x,y)) \vdash \quad \vdash \exists u \forall w (Q(u,w) \supset P(w)) \vdash C}{\vdash \forall x (P(x) \supset \exists y Q(x,y)) \vee \exists u \forall w (Q(u,w) \supset P(w)) \vdash C} \vee_L + \vee_L
 \end{array}$$



$$\begin{array}{c}
 \frac{\Sigma', y; T', P(t), Q(x,y) \vdash C'}{\Sigma', T', P(t) \vdash C'} \wedge_L \\
 \vdots \\
 \frac{\Sigma', T' \vdash C}{\vdash C} \\
 \frac{\Sigma', u; T'', Q(u,t), P(t) \vdash C''}{\Sigma', u; T'' \vdash C''} \wedge_L \\
 \vdots \\
 \frac{\Sigma', u; T'' \vdash C}{\vdash C} \\
 \frac{\Sigma', T' \vdash C \quad \Sigma', u; T'' \vdash C}{\vdash C} \vee
 \end{array}$$

We can use the 2-system to derive the axiom:

$$\begin{array}{c}
 \frac{x'; P(x'), Q(x', x') \vdash Q(x', x')}{x'; P(x') \vdash Q(x', x')} \text{ } b_1 \\
 \frac{x'; P(x') \vdash Q(x', x')}{x'; P(x') \vdash \exists y Q(x', y)} \exists_e \\
 \frac{x'; P(x') \vdash \exists y Q(x', y)}{x'; \vdash P(x') \supset \exists y Q(x', y)} \supset_e \\
 \frac{x'; \vdash P(x') \supset \exists y Q(x', y)}{\vdash \forall x (P(x) \supset \exists y Q(x, y))} \forall_e \\
 \frac{\vdash \forall x (P(x) \supset \exists y Q(x, y)) \vee \exists u \forall w (Q(u, w) \supset P(w))}{\vdash \forall x (P(x) \supset \exists y Q(x, y)) \vee \exists u \forall w (Q(u, w) \supset P(w))} \vee_e \\
 \frac{u', w'; Q(u', w'), P(w') \vdash P(w')}{u', w'; Q(u', w') \vdash P(w')} \text{ } t_2 \\
 \frac{u', w'; Q(u', w') \vdash P(w')}{u', w' \vdash Q(u', w') \supset P(w')} \supset_e \\
 \frac{u' \vdash \forall w (Q(u, w) \supset P(w))}{u' \vdash \exists u \forall w (Q(u, w) \supset P(w))} \exists_e \\
 \frac{u' \vdash \exists u \forall w (Q(u, w) \supset P(w))}{u' \vdash \forall x (P(x) \supset \exists y Q(x, y)) \vee \exists u \forall w (Q(u, w) \supset P(w))} \forall_e \\
 \frac{u' \vdash \forall x (P(x) \supset \exists y Q(x, y)) \vee \exists u \forall w (Q(u, w) \supset P(w))}{\vdash \forall x (P(x) \supset \exists y Q(x, y)) \vee \exists u \forall w (Q(u, w) \supset P(w))} b
 \end{array}$$

We can translate the 2-system into the rule:

$$\frac{G \mid \Sigma', y; \Gamma', P(t), Q(t, y) \vdash C' \quad G \mid \Sigma'', u; \Gamma'', Q(u, t'), P(t') \vdash C''}{G \mid \Sigma'; \Gamma', P(t) \vdash C' \mid \Sigma'', u; \Gamma'', Q(u, t') \vdash C''}$$

and translate the derivation:

$$\begin{array}{c}
 \frac{x'; P(x'), Q(x', x') \vdash Q(x', x')}{x'; P(x') \vdash Q(x', x')} \quad \frac{u', w'; Q(u', w'), P(w') \vdash P(w')}{u', w'; Q(u', w') \vdash P(w')} \\
 \frac{x'; P(x') \vdash Q(x', x')}{x'; P(x') \vdash \exists y Q(x', y)} \quad \frac{u', w'; Q(u', w') \vdash P(w')}{u', w' \vdash Q(u', w') \supset P(w')} \\
 \frac{x'; P(x') \vdash \exists y Q(x', y)}{x' \vdash P(x') \supset \exists y Q(x, y)} \quad \frac{u', w' \vdash Q(u', w') \supset P(w')}{u' \vdash \forall w (Q(u, w) \supset P(w))} \\
 \frac{x' \vdash P(x') \supset \exists y Q(x, y)}{\not\vdash \forall x (P(x) \supset \exists y Q(x, y))} \quad \frac{u' \vdash \forall w (Q(u, w) \supset P(w))}{u' \vdash \exists u \forall w (Q(u, w) \supset P(w))} \\
 \frac{\not\vdash \forall x (P(x) \supset \exists y Q(x, y))}{\not\vdash \forall x (P(x) \supset \exists y Q(x, y)) \vee \exists u \forall w (Q(u, w) \supset P(w))} \quad \frac{u' \vdash \exists u \forall w (Q(u, w) \supset P(w))}{u' \vdash \forall x (P(x) \supset \exists y Q(x, y)) \vee \exists u \forall w (Q(u, w) \supset P(w))} \\
 \frac{\not\vdash \forall x (P(x) \supset \exists y Q(x, y)) \vee \exists u \forall w (Q(u, w) \supset P(w))}{\Sigma; \forall x (P(x) \supset \exists y Q(x, y)) \vee \exists u \forall w (Q(u, w) \supset P(w))} \text{ } \textcircled{?} \text{ } \text{EC}
 \end{array}$$

What happens to Σ in the last step?

Example 2: $\forall y \forall w (\exists x (P(x,y) \supset Q(x,y)) \vee \exists z (Q(z,w) \supset P(z,w)))$

$$\begin{array}{c}
 \begin{array}{c}
 \frac{x', Q \uparrow \vdash \uparrow}{x' \Downarrow Q(x', y) \vdash} \quad \frac{}{x' \vdash P(x', y) \Downarrow} \vdash \\
 \hline
 x' \Downarrow P(x', y) \supset Q(x', y) \vdash \\
 \vdots \\
 \frac{x' \uparrow P(x', y) \supset Q(x', y) \vdash \uparrow}{\uparrow \exists x (\quad) \vdash \uparrow} \\
 \hline
 \uparrow \exists x (P(x, y) \supset Q(x, y)) \vee \exists z (Q(z, w) \supset P(z, w)) \vdash \\
 \hline
 \Downarrow \forall y \forall w (\quad) \vdash
 \end{array}
 \end{array}$$

$$\frac{\Sigma_1, x; P, Q, T_1 \Rightarrow \Delta_1}{\Sigma_1, x; P, T_1 \Rightarrow \Delta_1} \quad t_1$$

$$\frac{\Sigma_2, z; Q, P, T_2 \Rightarrow \Delta_2}{\Sigma_2, z; Q, T_2 \Rightarrow \Delta_2} \quad t_2$$

$$\frac{\vdots \quad \vdots}{\Sigma, x; T \Rightarrow \Delta \quad \Sigma, z; T \Rightarrow \Delta} \quad \vdash \\
 \hline
 \Sigma, T \vdash \Delta$$

and the rule:

$$\frac{\Sigma_1, x; P(x, y), Q(x, y), T_1 \Rightarrow \Delta_1 \quad \Sigma_2, z; Q(z, w), P(z, w), T_2 \Rightarrow \Delta_2}{\Sigma_1, x; P(x, y), T_1 \Rightarrow \Delta_1 \mid \Sigma_2, z; Q(z, w), T_2 \Rightarrow \Delta_2}$$

$$\begin{array}{c}
\frac{\gamma', \omega', x', x''; P, Q \vdash P}{\gamma', \omega', x', x''; P \vdash Q} \quad t_1 \qquad \frac{\gamma', \omega', z', z''; Q, P \vdash P}{\gamma', \omega', z', z''; Q \vdash P} \quad t_2 \\
\frac{\gamma', \omega', x', x''; P \vdash Q}{\gamma', \omega', x'; \vdash P_{x''} \supset Q_{x''}} \\
\frac{\gamma', \omega', x'; \vdash \exists x (P \supset Q)}{\gamma', \omega', x'; \vdash \exists x P_{x''} \supset Q_{x''} \vee \exists z Q_{z''} \supset P_{z''}} \\
\frac{\gamma', \omega', x'; \vdash \exists x P_{x''} \supset Q_{x''} \vee \exists z Q_{z''} \supset P_{z''}}{\gamma', \omega', x'; \vdash \exists x (P_{x''} \supset Q_{x''}) \vee \exists z (Q_{z''} \supset P_{z''})} \quad \vee^2 \\
\frac{\gamma', \omega', x'; \vdash \exists x (P_{x''} \supset Q_{x''}) \vee \exists z (Q_{z''} \supset P_{z''})}{\vdash \forall y \forall \omega (\exists x (P \supset Q) \vee \exists z (Q \supset P))}
\end{array}$$

translates to:

$$\begin{array}{c}
\frac{\gamma', \omega', x', x''; P(x'', y'), Q(x'', y') \vdash Q(x'', y')}{\gamma', \omega', x', x''; P(x'', y') \vdash Q(x'', y')} \quad \gamma', \omega', z', z''; P(z'', w'), Q(z'', w') \vdash P(z'', w') \\
\frac{\gamma', \omega', x', x''; P(x'', y') \vdash Q(x'', y') \mid \gamma', \omega', z', z''; Q(z'', w') \vdash P(z'', w')}{\gamma', \omega', x', x''; \vdash P(x'', y') \supset Q(x'', y') \mid \gamma', \omega', z', z''; Q(z'', w') \vdash P(z'', w')} \\
\frac{\gamma', \omega', x'; \vdash \exists x (P(x, y') \supset Q(x, y')) \mid \gamma', \omega', z', z''; Q(z'', w') \vdash P(z'', w')}{\gamma', \omega', x'; \vdash \exists x (P(x, y') \supset Q(x, y')) \mid \gamma', \omega', z', z''; \vdash Q(z'', w') \supset P(z'', w')} \\
\frac{\gamma', \omega', x'; \vdash \exists x (P(x, y') \supset Q(x, y')) \mid \gamma', \omega', z', z''; \vdash Q(z'', w') \supset P(z'', w')}{\gamma', \omega', x'; \vdash \exists x (P(x, y') \supset Q(x, y')) \mid \gamma', \omega', z', z''; \vdash \exists z (Q(z, w') \supset P(z, w'))} \\
\frac{\gamma', \omega', x'; \vdash \exists x (P(x, y') \supset Q(x, y')) \vee \exists z (Q(z, w') \supset P(z, w')) \mid \gamma', \omega', z'; \vdash \exists x (P(x, y') \supset Q(x, y')) \vee \exists z (Q(z, w') \supset P(z, w'))}{\gamma', \omega' \vdash \exists x (P(x, y') \supset Q(x, y')) \vee \exists z (Q(z, w') \supset P(z, w'))} \\
\vdash \forall y \forall \omega (\exists x (P(x, y) \supset Q(x, y)) \vee \exists z (Q(z, w) \supset P(z, w)))
\end{array}$$

NOTE: I feel like there is something weird w/ the example, concerning the Σ . I tried to do it anyway but something feels wrong...



the bottom rule adds x' and z' to each component

+ $\overline{EC}$